



2.15

Semi-log Plots

Student Notes and Guided Practice

AP Precalculus - Unit 2

| ECHS Mathematics

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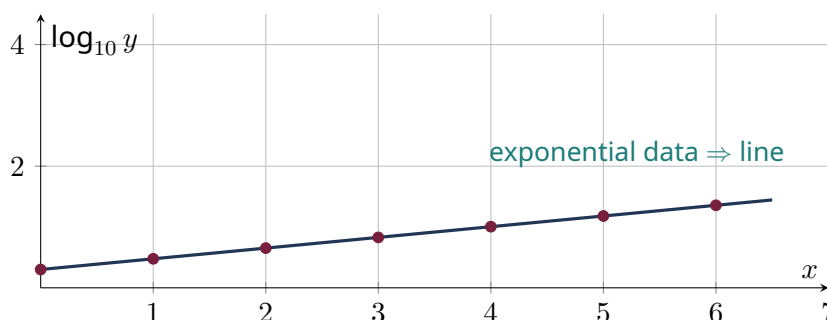
Ghuwaleh

Learning Targets

- Interpret semi-log plots with a logarithmic output axis.
- Use line slope and intercept to construct exponential models.
- Explain how base choice and vertical shifts affect linearization.

Essential Question

Why does taking a logarithm of exponential output data produce a line?



Concept Framework

Key Idea

If $y = ab^x$, then $\log y = \log a + x \log b$.

Key Idea

On a log-output plot, slope is $\log b$ and intercept is $\log a$.

Key Idea

With base 10, $a = 10^{\text{intercept}}$, $b = 10^{\text{slope}}$.

Key Idea

A vertical shift $y = k + ab^x$ is not linearized by plotting $\log y$; plot $\log(y - k)$ instead.

Formula Map**Formula and Structure**

$$y = ab^x \Rightarrow \log_{10} y = \log_{10} a + x \log_{10} b$$

Formula and Structure

$$a = 10^c, b = 10^m$$

Worked Examples**Worked Example 1**

Problem. A semi-log line is $Y = 0.301 + 0.176x$, where $Y = \log_{10} y$. Find the model.

Solution. $a = 10^{0.301} \approx 2$, $b = 10^{0.176} \approx 1.5$. Thus $y \approx 2(1.5)^x$.

Worked Example 2

Problem. What does curvature on a semi-log plot indicate?

Solution. A single exponential model is not adequate over the full interval, or a shift/other structure has not been accounted for.

Worked Example 3

Problem. Why does changing the log base not change whether the transformed data are linear?

Solution. Different log bases multiply all transformed values by a constant, preserving linearity while rescaling slope/intercept.

Multiple-Representation Connection**AP Reasoning**

For this topic, always connect at least two representations: algebraic structure, numerical evidence, graphical behavior, or contextual meaning. State restrictions and units before interpreting a result. A correct calculation without a valid domain or contextual interpretation is incomplete AP reasoning.

Common Errors to Avoid

Common Error Check

- Do not discard domain restrictions when rewriting an expression.
- Do not infer local behavior from only one interval-wide average.
- Do not report a numerical answer without units or contextual meaning when quantities are named.
- Do not use a graphing window as proof that a feature does not exist.

Guided Check

1. A base-10 semi-log line is $Y = 0.301 + 0.176x$, where $Y = \log_{10} y$. Find the exponential model.

2. A base-10 semi-log line is $Y = 1 + 0.301x$, where $Y = \log_{10} y$. Find the exponential model.

3. A base-10 semi-log line is $Y = 0 + -0.097x$, where $Y = \log_{10} y$. Find the exponential model.

4. A base-10 semi-log line is $Y = 2 + 0.4771x$, where $Y = \log_{10} y$. Find the exponential model.

5. A base-10 semi-log line is $Y = -1 + 0.2x$, where $Y = \log_{10} y$. Find the exponential model.

6. An exponential has $a=5, b=1.2$. Find semi-log slope/intercept in base 10.

Lesson Summary

Complete these sentences before leaving the lesson:

1. The most important structure in this topic is _____.
2. A restriction I must remember is _____.
3. One graphical feature I can justify algebraically is _____.